EXCEL & POWER BI FORMULA REFERENCE

South Indian Bank – Business Performance & Regional Analysis

Formula documentation, calculation logic and project workflow

Purpose of this document

This document records the calculation logic relevant to the Excel workbooks and the Power BI analysis. It is based on the two supplied workbooks: “REGIONAL WISE ANALYSIS.xlsx” and “SIB BRANCH WISE BUSINESS POSITION AND TARGET VS ACHIEVEMENT.xlsx”. The regional workbook contains visible Excel formulas in the Gap/Excess columns and SMA Reduction sheet. The branch-wise workbook contains pre-calculated values rather than embedded Excel formulas. Therefore, formulas shown for the branch-wise workbook below are reproducible formula examples, not claims that those formulas were present in the supplied file.

Excel formulas identified in the supplied Regional Wise Analysis workbook

The following formulas were found in the workbook. They are repeated down the 19 regional-office records in the relevant sheets.

Worksheet	Column / Field	Excel formula (example row 2)	Meaning
TOTAL BUSINESS	Gap / Excess (Column J)	=F2-I2	Actual business as on 13.02.2026 minus Feb-26 target. Positive = above target; negative = shortfall.
RETAIL TERM DEPOSIT	Gap / Excess (Column J)	=F2-I2	Actual retail term deposits minus target.
CASA EOD	Gap / Excess (Column J)	=F2-I2	Actual CASA EOD minus target.
SB EOD	Gap / Excess (Column J)	=F2-I2	Actual SB EOD minus target.
CD EOD	Gap / Excess (Column J)	=F2-I2	Actual CD EOD minus target.
CASA AMB	Gap / Excess (Column J)	=F2-I2	Actual CASA average monthly balance minus target.
SB AMB	Gap / Excess (Column J)	=F2-I2	Actual SB average monthly balance minus target.
CD AMD	Gap / Excess (Column J)	=F2-I2	Actual CD average monthly balance minus target; worksheet label is “CD AMD”.
STANDARD ADVANCE	Gap / Excess (Column J)	=F2-I2	Actual standard advances minus target.
SMA REDUCTION	Gap Excess (Column F)	=E2-D2	January target minus actual SMA reduction value. Positive = target exceeds actual; interpret with the bank’s SMA convention.

Excel calculation formulas for the branch-wise business dataset

In the supplied branch-wise workbook, the “Data Set” sheet has field headings on row 3 and branch records beginning on row The following formulas can be used to independently reproduce or validate the MTD, YTD and target-gap calculations. The column letters below refer to the visible layout of that supplied worksheet.

Metric	Example Excel formula (row 4)	Calculation / interpretation
Total Business – MTD Change	=I4-H4	13.02.2026 total business less 31.01.2026 total business.
Total Business – YTD Change	=I4-G4	13.02.2026 total business less 31.03.2025 total business.
Total Business – Gap / Excess	=I4-L4	13.02.2026 actual total business less Feb-26 target.
Retail Term Deposit – MTD	=P4-O4	Current reported value less previous month-end value.
Retail Term Deposit – YTD	=P4-N4	Current reported value less 31.03.2025 value.
Retail Term Deposit – Gap	=P4-S4	Current reported value less target.
CASA EOD – MTD	=W4-V4	Current reported CASA EOD less previous month-end value.
CASA EOD – YTD	=W4-U4	Current reported CASA EOD less 31.03.2025 value.
CASA EOD – Gap	=W4-Z4	Current reported CASA EOD less target.
Gold Loan – MTD	=DO4-DN4	13.02.2026 Gold Loan value less 31.01.2026 value.
Gold Loan – YTD	=DO4-DM4	13.02.2026 Gold Loan value less 31.03.2025 value.
Gold Loan – Gap / Excess	=DO4-DR4	Current Gold Loan value less Feb-26 target.
Achievement % (additional KPI)	=IFERROR(I4/L4,0)	Actual total business divided by target. Format the result as Percentage.

Note: The branch-wise workbook already contains MTD Change, YTD Change and Gap/Excess values in many columns. The formulas above are provided as validation/recalculation logic. Confirm that the relevant actual and target columns are populated before applying them, and fill down for the remaining branch records.

Useful Excel formulas for data preparation and analysis

Purpose	Formula example	Use
Sum of a metric	=SUM(I4:I954)	Total the current-period value for all branch records.
Average achievement	=AVERAGE(IFERROR(I4:I954/L4:L954,0))	Average of row-level achievement ratios; in older Excel, this may require array handling.
Highest regional / branch value	=MAX(I4:I954)	Find the highest value in the selected metric column.
Lowest regional / branch value	=MIN(I4:I954)	Find the lowest value in the selected metric column.
Rank by business	=RANK.EQ(I4,$I$4:$I$954,0)	Rank branches from highest to lowest current business.

Conditional status	=IF(I4>=L4,"Target Achieved","Below Target")	Classify a branch based on actual versus target.
Regional total	=SUMIF($D$4:$D$954,A2,$I$4:$I$954)	Sum current business for the region named in A2.
Regional business share	=IFERROR(B2/SUM($B$2:$B$20),0)	Regional business divided by the sum of regional business. Format as Percentage.

Power BI – recommended DAX measures

The supplied files do not include a Power BI (.pbix) file or exported DAX measure definitions. Consequently, the measures below are recommended DAX implementations based on the fields and analyses described in the project documentation; they are not verified extracts of measures already created in your Power BI report.

Before creating the measures, use Power Query to rename the imported columns to simple names such as Regional Office, Total Business Actual, Total Business Target, MTD Change, YTD Change, Gold Loan Actual, Gold Loan Target and SMA Reduction Actual. The examples assume the main table is named 'Data Set'. Adjust names to match your actual Power BI model.

Measure name	DAX formula	Purpose
Total Business Actual	Total Business Actual = SUM('Data Set'[Total Business Actual])	Sum of current total business across the selected branches/regions.
Total Business Target	Total Business Target = SUM('Data Set'[Total Business Target])	Sum of assigned total business targets.
Total Business Gap	Total Business Gap = [Total Business Actual] - [Total Business Target]	Positive means achievement exceeds target; negative means a shortfall.
Achievement %	Achievement % = DIVIDE([Total Business Actual], [Total Business Target], 0)	Actual business as a proportion of target. Format as Percentage.
Total MTD Growth	Total MTD Growth = SUM('Data Set'[MTD Change])	Aggregate month-to-date change.
Total YTD Growth	Total YTD Growth = SUM('Data Set'[YTD Change])	Aggregate year-to-date change.
Gold Loan Actual	Gold Loan Actual = SUM('Data Set'[Gold Loan Actual])	Current Gold Loan business value.
Gold Loan Target	Gold Loan Target = SUM('Data Set'[Gold Loan Target])	Gold Loan target value.
Gold Loan Gap	Gold Loan Gap = [Gold Loan Actual] - [Gold Loan Target]	Difference between actual Gold Loan value and target.
Gold Loan Achievement %	Gold Loan Achievement % = DIVIDE([Gold Loan Actual], [Gold Loan Target], 0)	Gold Loan target achievement ratio.
Regional Business Share	Regional Business Share % = DIVIDE([Total Business Actual], CALCULATE([Total Business Actual], ALL('Data Set'[Regional Office])), 0)	Share of total business attributable to each region in the current filter context. Format as Percentage.
Branch Count	Branch Count = DISTINCTCOUNT('Data Set'[Sol ID])	Number of unique branches in the current filter context.

Suggested DAX for SMA reduction

Measure name	DAX formula	Interpretation
SMA Reduction Actual	SMA Reduction Actual = SUM('SMA Reduction'[Actual SMA Reduction])	Sum of actual SMA reduction values, if the SMA sheet is loaded as a separate table.
SMA Reduction Target	SMA Reduction Target = SUM('SMA Reduction'[SMA Target])	Sum of SMA reduction targets.
SMA Target Gap	SMA Target Gap = [SMA Reduction Target] - [SMA Reduction Actual]	Target less actual, matching the sign convention used by the Excel SMA sheet.

For the SMA measures, confirm the source column names and whether a higher or lower value represents better performance. The Excel source calculates SMA Gap/Excess as Target minus Actual (E minus D), which is the reverse sign convention from the business target-gap formula (Actual minus Target).

How to create the project in Power BI – step-by-step

Step 1 – Import data: Open Power BI Desktop → Home → Get Data → Excel Workbook. Select the regional analysis workbook and branch-wise business workbook. Select the required sheets, especially TOTAL BUSINESS, GOLD LOAN, SMA REDUCTION and Data Set.

Step 2 – Transform data: Choose Transform Data. In Power Query, promote headers where required, remove blank rows, set numeric fields to Decimal Number, set codes/IDs to the appropriate data type, trim/clean text fields and standardise Regional Office names.

Step 3 – Rename fields: Rename long or repeated column labels to clear names. For example: Total Business Actual, Total Business Target, MTD Change, YTD Change, Gold Loan Actual and Gold Loan Target. Keep the source date in the field name or add a separate reporting-date field to avoid confusion.

Step 4 – Check data quality: Check nulls, duplicate branch IDs, inconsistent region spellings and unexpected negative values. Do not remove legitimate negative MTD/YTD movements; they may represent a decline.

Step 5 – Build the model: If you load multiple tables, create a distinct Regional Office dimension and relate it one-to-many to the relevant fact tables. Use a unique branch/SOL ID for branch-level relationships. Avoid many-to-many relationships unless justified.

Step 6 – Create DAX measures: Select the relevant table → Modeling → New measure. Add the DAX measures listed in Sections 5 and 6. Format percentage measures as Percentage and monetary/business values using the unit shown in the source.

Step 7 – Create visuals: Use clustered bar/column charts for target versus achievement; horizontal bars for top 10 MTD/YTD branches; cards for actual, target, gap and achievement percentage; and a bar chart for regional business share.

Step 8 – Add slicers: Add Regional Office, Branch Name and product/parameter slicers where the data model supports them. Confirm that slicers filter all intended visuals.

Step 9 – Validate: Compare Power BI totals with Excel totals for the same sheet, reporting date and filter selection. Check a few individual branches manually before finalising the dashboard.

Step 10 – Save and document: Save the .pbix file. Export dashboard screenshots and include the data source, reporting date, calculation definitions and limitations in the project documentation.

Suggested dashboard pages and visuals

Dashboard page	Recommended visuals	Key question answered
Executive Overview	KPI cards: Total Business Actual, Target, Gap, Achievement %; regional ranking bar chart	What is the overall business position?
Regional Performance	Clustered bar chart: actual vs target by Regional Office; regional share bar chart	Which regions have the highest business volume and achievement?
Branch Growth	Top 10 MTD and Top 10 YTD horizontal bar charts	Which branches show the strongest recent and cumulative growth?
Product Performance	Gold Loan actual vs target; product-wise target-gap chart	How are key products performing against targets?
Asset Quality	SMA reduction actual vs target by region	How are regions progressing against SMA reduction targets?

Important interpretation and data limitations

- The branch-wise source workbook contains values for many calculated fields; the workbook does not contain embedded formulas in its Data Set sheet. The Excel formulas in Section 3 are reproducible calculation logic.
- The regional workbook includes formulas for Gap/Excess in several sheets and for SMA Reduction. These are the formulas directly identified in the uploaded file.
- Business volume, target achievement and growth are not the same as net profit. Regional business share must not be described as regional profit contribution unless region-wise income, funding cost, operating expenses, credit cost and provisions are available.
- The available business snapshot is dated 13 February 2026 (with targets labelled February 2026). It should not be represented as a full-year FY 2025–26 closing regional position.
- The DAX expressions in this document are templates. Update the table and column names to match the final Power BI model and validate every measure against the Excel source.

Suggested methodology paragraph for the project report

The project involved importing the branch-wise and regional banking datasets into Microsoft Excel for review and calculation validation, followed by data transformation and visual analysis in Microsoft Power BI. Month-to-date movement was calculated as the current reporting value less the previous month-end value, while year-to-date movement was calculated as the current reporting value less the 31 March 2025 base value. Target gap was calculated as actual business less the assigned target, except for the SMA Reduction worksheet, where the source workbook uses target less actual. DAX measures were used or can be used to aggregate actuals, targets, gaps, achievement percentages and regional business shares

dynamically according to report filters. The resulting dashboards support comparison of regional and branch performance and identification of areas requiring management attention.

Appendix – source workbook fields referenced

Regional workbook sheets reviewed: TOTAL BUSINESS, RETAIL TERM DEPOSIT, CASA EOD, SB EOD, CD EOD, CASA AMB, SB AMB, CD AMD, STANDARD ADVANCE, GOLD LOAN and SMA REDUCTION. Branch-wise workbook sheets reviewed: Data Set and Core Parameter Details. The formulas and examples in this document are mapped to the supplied workbook layouts.